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Introduction

Background and Motivation

Optimization without deep learning

- The revolutionary paper, *Problem Complexity and Method Efficiency in Optimization*, is published in 1983.

The Multiplicative Revolution

- Another revolutionary paper, *The Exponentiation Gradient Descent Algorithm* by J. Kivinen and M. Warmuth, is published in 1997.

The Conflict

- *Understanding deep learning requires rethinking generalization* (2017) by Zhang et al. is published in ICLR 2017.
- **The Core Mystery:** Deep neural networks can easily fit random labels, challenging traditional VC-dimension and generalization theories.
- Out of infinite solutions that interpolate the data, why does SGD find the ones that generalize?

Problem Setup and Notation

Overparameterized Nonlinear Models

- **Dataset:** $\{(x_i, y_i)\}_{i=1}^n$ where $x_i \in \mathbb{R}^d$ and $y_i \in \mathbb{R}$.
- **Model:** $f(x_i, w) = f_i(w)$ with parameters $w \in \mathbb{R}^p$.
- **The Interpolating Regime:** $p \gg n$.

The Set of Global Minima (Interpolants)

The set of all parameters consistent with the training data:

$$\mathcal{W} = \{w \in \mathbb{R}^p \mid f_i(w) = y_i, \quad i = 1, \ldots, n\} \tag{1}$$

This forms a high-dimensional $(p - n)$-dimensional manifold.

The Learning Objective

Minimize empirical risk $L(w) = \sum_{i=1}^n L_i(w)$, where $L_i(w) = \ell(y_i, f_i(w))$. Every $w \in \mathcal{W}$ yields $L(w) = 0$.

Optimization via SGD

Standard training typically utilizes **Stochastic Gradient Descent (SGD)**:

$$w_i = w_{i-1} - \eta \nabla L_i(w_{i-1}), \quad i \geq 1 \tag{2}$$

- $L_i(\cdot)$ is differentiable and non-negative.
- $L_i(w) = 0$ iff $y_i = f_i(w)$.
- For $i > n$, we cycle through the data or select at random.

How does this update rule interact with the geometry of $\mathcal{W}$?

Stochastic Mirror Descent (SMD)

The Geometry of Mirror Descent

Instead of updating in the weight space, SMD updates in the **mirrored domain** via a strictly convex potential function $\psi(\cdot)$.

The Update Rule

The gradient $\nabla\psi(\cdot)$ defines an invertible map. The recursion is:

$$\nabla\psi(w_i) = \nabla\psi(w_{i-1}) - \eta\nabla L_i(w_{i-1}) \tag{3}$$

- **Intuition:** The update happens in the "dual" space determined by $\nabla\psi$.
- **Special Case (SGD):** If $\psi(w) = \frac{1}{2}\|w\|^2$, then $\nabla\psi$ is the identity map, and we recover standard SGD.
- **Choice of Potential:** Different ψ exploit different geometric structures (e.g., sparsity, simplex).

The Bregman Perspective

The SMD update can be equivalently viewed as a proximal minimization problem:

$$w_i = \arg\min_{w} \left\{ \eta w^T \nabla L_i(w_{i-1}) + D_\psi(w, w_{i-1}) \right\} \tag{4}$$

Bregman Divergence $D_\psi(w, \bar{w})$

Defined as the difference between the function and its first-order Taylor expansion:

$$D_\psi(w, \bar{w}) := \psi(w) - \psi(\bar{w}) - \nabla\psi(\bar{w})^T (w - \bar{w}) \tag{5}$$

- **Non-negativity:** $D_\psi(w, \bar{w}) \geq 0$ and $D_\psi(w, \bar{w}) = 0$ iff $w = \bar{w}$.
- **Distance Metric:** It acts as a "proximity measure" tailored to the geometry of the potential ψ.

Implicit Regularization in Linear Models

Consider the overparameterized linear model $f(x_i, w) = x_i^T w$.

Proposition 1: Convergence to the Closest Interpolant

For a sufficiently small step size η, the SMD iterates converge to:

$$w_\infty = \arg\min_{w \in \mathcal{W}} D_\psi(w, w_0) \tag{6}$$

- **The Result:** Out of all infinite solutions in $\mathcal{W}$, SMD picks the one "closest" to the initialization in the Bregman sense.
- **Corollary (Initialization):** If we initialize at the global minimum of ψ, then:

$$w_\infty = \arg\min_{w \in \mathcal{W}} \psi(w) \tag{7}$$

- **Conclusion:** Optimization geometry **is** the regularizer.

Assumption 1: Local Solvability and Restricted Convexity

The analysis considers a local region $\mathcal{B} = \{w \in \mathbb{R}^p \mid D_\psi(w, \bar{w}) \leq \epsilon\}$ containing the initialization w_0.

Condition on the Loss Function

There exists $\bar{w} \in \mathcal{W}$ such that for all $w \in \mathcal{B}$ and $i = 1, \ldots, n$:

$$D_{L_i}(\bar{w}, w) = L_i(\bar{w}) - L_i(w) - \nabla L_i(w)^T(\bar{w} - w) \geq 0 \tag{8}$$

- Since $L_i(\bar{w}) = 0$ is a global minimum, this requires the loss to be locally "star-convex" toward the solution $\bar{w}$.
- This assumption is consistent with the behavior of overparameterized models near the interpolating manifold.

Assumption 2: Regularity of the Model Function

The convergence proof requires the model $f_i(w)$ to exhibit bounded first and second-order derivatives within the convex hull of $\mathcal{B}$.

- **Gradient Bound:** $\|\nabla f_i(w)\| \leq \gamma$ for all $w \in \text{conv}(\mathcal{B})$.
- **Hessian Stability:** There exist constants α and β such that:

$$\alpha I \preceq H_{f_i}(w) \preceq \beta I \tag{9}$$

  for all $i = 1, \ldots, n$ and $w \in \text{conv}(\mathcal{B})$.

These bounds ensure that the nonlinearity of $f(\cdot)$ does not lead to divergence under a fixed learning rate η.

Theorem 3: Deterministic Convergence

Under Assumption 1, and for a step size η such that $\psi(\cdot) - \eta L_i(\cdot)$ is strictly convex for all i, the following hold for the SMD iterates $\{w_i\}$:

1. **Containment:** All iterates $\{w_i\}$ remain within the region $\mathcal{B}$.
2. **Global Convergence:** The sequence $\{w_i\}$ converges to a limit w_∞.
3. **Interpolation:** The limit point is a global minimum, i.e., $w_\infty \in \mathcal{W}$.

Note: Unlike many deep learning results, this convergence is *deterministic* and does not rely on expectation or high-probability bounds.

Theorem 4: Characterization of the Limit Point

Let $w^* = \arg\min_{w \in \mathcal{W}} D_\psi(w, w_0)$ be the Bregman-closest interpolant to the initialization.

Proximity to the Optimal Regularized Solution

Under Assumptions 1 and 2, the limit point w_∞ satisfies:

1. $D_\psi(w_\infty, w_0) = D_\psi(w^*, w_0) + o(\epsilon)$
2. $D_\psi(w^*, w_\infty) = o(\epsilon)$

If w_0 is $O(\epsilon)$ from the manifold $\mathcal{W}$, then w_∞ is $o(\epsilon)$ from w^*. This establishes that SMD approximately minimizes the potential $\psi(\cdot)$ even in the nonlinear setting.

Lemma 6: A General Identity for SMD

The authors derive a fundamental identity that holds for any model $f(\cdot, \cdot)$ and differentiable loss $\ell(\cdot)$. For any $w \in \mathcal{W}$:

$$D_\psi(w, w_{i-1}) = D_\psi(w, w_i) + D_{\psi - \eta L_i}(w_i, w_{i-1}) + \eta L_i(w_i) + \eta D_{L_i}(w, w_{i-1}) \tag{10}$$

- **Significance:** If the step size is small enough to keep $D_{\psi - \eta L_i}$ positive, then $D_\psi(w, w_i)$ is non-increasing.
- **Link to H-∞ Control:** This identity relates SMD to minimax optimality and robust estimation theory developed in the 1990s.
- **Consequence:** It provides a direct Lyapunov-like argument for convergence in Bregman space.

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Core Contributions and Impact

- **Formalization of Implicit Bias:** Provides an exact (linear) and approximate (nonlinear) characterization of the solutions found by Mirror Descent.
- **Extension to Nonlinearity:** Uses local regularity assumptions (RSC, Star-convexity) to bridge the gap between linear theory and deep learning.
- **Optimizer as Regularizer:** Demonstrates that the choice of mirror map ψ determines the inductive bias (e.g., L_1 vs. L_∞), impacting generalization on datasets like CIFAR-10.
- **Algorithmic Framework:** Establishes a unified perspective on SGD, Exponentiated Gradient, and p-norm algorithms as interpolating solvers with specific geometric properties.